THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHS · EASY LEVEL FULL-YEAR PAPER SET 3

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. For the data with classes 20–30, 30–40, 40–50, 50–60, 60–70 with frequencies 7, 17, 20, 3, 3 respectively, The median class is: 1

(a) 20–30 **(b)** 30–40 **(c)** 50–60 **(d)** 40–50

2. A coin is tossed two times. The probability of getting two heads is: 1

(a) $\frac{3}{4}$ **(b)** $\frac{1}{4}$ **(c)** $\frac{1}{2}$ **(d)** 0

3. Which of the following is an irrational number? 1

(a) $\frac{5}{8}$ **(b)** 25 **(c)** $\sqrt{7}$ **(d)** $\sqrt{25}$

4. The product of the zeroes of $x^2 - 5x + 6$ is: 1

(a) 11 **(b)** 6 **(c)** 5 **(d)** -6

5. The sum of two numbers is 10 and their difference is 4. The larger number is: 1

(a) 3 **(b)** 14 **(c)** 6 **(d)** 7

6. The roots of $(x - 2)(x - 3) = 0$ are: 1

(a) 2 and 3 **(b)** -2 and -3 **(c)** 2 and -3 **(d)** 5 and 6

7. Which of the following sequences is an AP? 1

(a) $4, 7, 13, 22, \ldots$ **(b)** $4, 7, 11, 15, \ldots$ **(c)** $4, 7, 10, 13, \ldots$ **(d)** $4, 7, 13, 16, \ldots$

8. The areas of two similar triangles are in the ratio 4:9. The ratio of their corresponding sides is 2: 1

(a) 5 **(b)** 9 **(c)** 6 **(d)** 3

9. The coordinates of the point on the x-axis at distance 3 from the origin on the right are: 1

(a) $(0,\ 3)$ **(b)** $(-3,\ 0)$ **(c)** $(3,\ 0)$ **(d)** $(3,\ 3)$

10. The value of $\sin^2 20^\circ + \cos^2 20^\circ$ is: 1

(a) 2 **(b)** 0 **(c)** 20 **(d)** 1

11. A vertical pole is 9 m high. The angle of elevation of the sun is 45° and the pole casts a shadow on level ground. The length of the shadow is: 1

(a) 18 m **(b)** $\frac{9}{2}$ m **(c)** 9 m **(d)** $9\sqrt{3}$ m

12. Tangents PA and PB are drawn from an external point P to a circle of centre O and radius 3 cm. The length of each tangent is 4 cm. The perimeter of the quadrilateral $PAOB$ is: 1

(a) 7 cm **(b)** 14 cm **(c)** 8 cm **(d)** 16 cm

13. The area of a circle is 154 cm^2. The radius of the circle is: 1

(a) 14 cm **(b)** 7 cm **(c)** 49 cm **(d)** 24.5 cm

14. A right circular cone has base radius 7 cm and height 24 cm. The curved surface area of the cone is: 1

(a) 704 cm² **(b)** 550 cm² **(c)** 528 cm² **(d)** 154 cm²

15. For the data with classes 20–30, 30–40, 40–50, 50–60, 60–70 with frequencies 3, 9, 4, 18, 16 respectively, The total number of observations $\sum f_i$ is: 1

(a) 49 **(b)** 51 **(c)** 50 **(d)** 47

16. A bag contains 3 red, 5 blue balls. One ball is drawn at random. The probability that it is red is: 1

(a) $\frac{1}{2}$ **(b)** $\frac{5}{8}$ **(c)** $\frac{1}{4}$ **(d)** $\frac{3}{8}$

17. The prime factorisation of 60 is: 1

(a) 2×29 **(b)** 2×31 **(c)** 2^6 **(d)** $2^2 \times 3 \times 5$

18. A polynomial of degree 2 has at most how many zeroes? 1

(a) 2 **(b)** 1 **(c)** 3 **(d)** infinitely many

19. The graph of $x = 2$ is a line parallel to the: 1

(a) line $y = x$ **(b)** x-axis **(c)** y-axis **(d)** none of the axes

20. The discriminant of $x^2 - 5x + 6 = 0$ is: 1

(a) 1 **(b)** 49 **(c)** 13 **(d)** -1

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. The marks obtained by students in a test are given below. 2

Marks	10–20	20–30	30–40	40–50
Number of students	5	8	20	17

Find the mean.

22. A die is thrown once. Find the probability of getting a number greater than 6. 2

23. Express 140 as a product of prime factors. 2

24. Find the zero of the polynomial $3x - 12$. 2

25. Solve: $x + y = 12$, $x - y = 4$. 2

26. Find the roots of $(x - 4)(x - 5) = 0$. 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 40–60 words each.

27. Solve by the substitution method: $2x + 3y = 18$ and $x + y = 7$. 3

28. Find the discriminant of $2x^2 - 4x + 3 = 0$ and hence the nature of its roots. 3

29. Find the sum of the first 10 terms of the AP $2, 5, 8, 11, \ldots$ 3

30. In $\triangle ABC$, D and E lie on AB and AC with $DE \parallel BC$. If $AD = 2$ cm, $DB = 3$ cm and $AE = 4$ cm, find EC. 3

31. Find the midpoint of the segment joining $(1,\ 7)$ and $(9,\ 3)$. 3

OR

If $\sin A = \frac{4}{5}$, find $\cos A$ and $\tan A$. 3

32. From the top of a cliff $10\sqrt{3}$ m high, the angle of depression of a boat on the sea is $60°$. Find the horizontal distance of the boat from the foot of the cliff. 3

OR

Tangents PA and PB are drawn from an external point P to a circle of centre O and radius 3 cm. The length of each tangent is 4 cm. Find the area of the quadrilateral $PAOB$ (in cm^2). 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. In an AP, $a = 6$, $d = 4$, $n = 20$. Find a_n and S_n. 5

34. A pole 6 m high casts a shadow 4 m long. At the same time a tower casts a shadow 30 m long. Find the height of the tower. 5

35. Find the centroid of the triangle with vertices $(0, 0)$, $(6, 0)$ and $(0, 9)$. 5

36. In $\triangle PQR$, right-angled at Q, $PQ = 6$ cm and $QR = 8$ cm. Find $\sin P$, $\cos P$, $\tan P$ and check that $\sin^2 P + \cos^2 P = 1$. 5

OR

The angle of elevation of the top of a vertical tower from a point on level ground is $30°$. After walking 6 m towards the foot of the tower along a straight line, the angle of elevation becomes $60°$. Find the height of the tower and the initial distance of the point from its foot (in this order). 5

37. Two concentric circles have radii 25 cm and 7 cm. A chord of the larger circle touches the smaller circle. Find the length of the chord. 5

OR

The largest possible circle is cut from a square sheet of side 14 cm. Find the area of the sheet left over. 5

38. A solid metallic sphere of radius 6 cm is melted and recast into small solid spheres of radius 2 cm. Find the number of small spheres obtained. 5

OR

The distances (in km) travelled to office by employees of a company are given below. 5

Distance (km)	10–20	20–30	30–40	40–50	50–60
Number of employees	10	2	6	16	6

Find the median.

END OF QUESTION PAPER — **Total: 80 Marks**